A multivariate Bayesian framework for diachronic dating of ancient texts: Validation on Biblical Hebrew and Ancient Greek

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Abstract

Linguistic dating of ancient texts — inferring composition date from internal morphosyntactic evidence — is central to biblical studies, classical philology, and Near Eastern linguistics, yet it has rarely been formalized as a probabilistic inference problem. Existing approaches identify features that shift between early and late corpora and assign ordinal labels ("archaic," "late"), but they do not produce calibrated probability distributions over dates. We present a framework that addresses this gap.

The core model (MLE-MVN) fits an ordinary-least-squares regression per morphosyntactic feature against date, assembles the residual covariance with Tikhonov (ridge) regularization, and evaluates a multivariate normal log-likelihood over a fine date grid to obtain a proper Bayesian posterior. A companion hierarchical Bayesian model with variational inference (HB-VI) extends this with group-level shrinkage and Monte Carlo uncertainty propagation, yielding better-calibrated credible intervals and enabling soft register assignment.

Three confound-correction modules are introduced: (1) a *resistant model* using only four clause-level syntactic features below conscious editorial control, providing an archaism diagnostic; (2) a *genre correction* (Strategies B+D) that down-weights features driven by legal versus narrative register rather than by date; and (3) a *word n-gram model* using POS-tag and function-word sequence features as an independent grammatical-sequence dating check.

The framework is trained on 22 prophetic and post-exilic Hebrew texts with independent historical anchoring (760–167 BCE) drawn from the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database. Held-out validation on oracle Jeremiah chapters (never seen in training) returns MAP = 562 BCE with mean LBH score = 0.16, consistent with a 7th-century text. HB-VI achieves holdout MAE of 16.8 yr versus 134.2 yr for MLE-MVN on three chronologically anchored holdouts. Cross-language validation on a 63-text Ancient Greek corpus (five holdouts) recovers all MAP dates within ∼30 yr of established scholarly dates, demonstrating that the methodology is language-agnostic. All code and corpora are publicly available (see Data Availability).

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as "archaic" or "late" and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

A notable recent contribution by Faigenbaum-Golovin et al. [5] demonstrated that statistical analysis of lemma n-gram frequencies can reliably distinguish among the three major scribal corpora of the Hebrew Bible — the old layers of Deuteronomy (D), the Deuteronomistic History (DtrH), and the Priestly source (P) — achieving approximately 84% chapter-level attribution accuracy on a 50-chapter benchmark, with interpretable discriminating-lemma lists linking computational results to traditional philological categories. That study establishes the viability of quantitative computational methods for biblical Hebrew scholarship and demonstrates that linguistically distinct scribal schools leave detectable statistical fingerprints even in relatively short texts. Its focus, however, is *authorship attribution*: given a passage of disputed origin, which known scribal milieu produced it? The present paper addresses the complementary but methodologically distinct question of *diachronic dating*: not "which school wrote it?" but "when was it written?" These are related but separable questions that demand different statistical frameworks, different feature sets, and different validation strategies, as detailed below.

This paper addresses two methodological gaps that authorship-attribution approaches leave open. First, no existing probabilistic framework provides date posteriors with credible intervals for Biblical Hebrew texts. Qualitative assessments of individual features are genuinely informative, but they cannot be combined into a single coherent posterior without a model that propagates feature-level evidence into a joint probability distribution over dates. Second, no principled framework separates the temporal signal from two well-known confounds: *archaism* (a late composer deliberately employing archaic vocabulary and morphology) and *genre* (legal or narrative register producing feature distributions that mimic either earlier or later stages of the language independently of composition date).

The framework presented here makes the following contributions:

1. A morphosyntactic feature extraction pipeline operating on the BHSA/ETCBC database [6, 7], producing 35 features across three tiers (lexical, morphological, and clause-level syntactic), selected by Spearman correlation ($p < 0.30$) with 100% LOO sign-consistency.

2. The MLE-MVN Bayesian dating model: per-feature OLS regression, Tikhonov-regularized residual covariance, and posterior over a 500-point date grid with explicit uncertainty quantification.

3. The HB-VI model: hierarchical Bayesian regression with mean-field variational inference (MFVI) and ADAM optimization [8, 9], providing proper parameter uncertainty propagation and soft register assignment.

4. A resistant model using only four clause-level features below conscious editorial control, yielding an archaism diagnostic from the divergence between the full and resistant MAP dates.

5. Genre correction (Strategies B+D) that down-weights features according to their sensitivity to legal versus narrative register.

6. A word n-gram model (POS-tag and function-word bigrams/trigrams) as an independent grammatical-sequence dating instrument.

7. Prior sensitivity analysis distinguishing data-driven from prior-dominated HB-VI date estimates.

8. Cross-language validation on a 63-text Ancient Greek corpus.

The paper is organized as follows. The Background section reviews the CBH/LBH distinction and the methodological debates surrounding linguistic dating. The Data section describes the training corpus and feature extraction. The MLE-MVN and HB-VI sections present the two Bayesian models. The Prior Sensitivity Analysis section covers the three-mode sensitivity framework. The Archaism Detection section presents the resistant model and archaism diagnostic. The Genre Correction section describes Strategies B+D. The Validation section reports holdout and cross-language results. The Discussion section addresses limitations, extensibility, and the relationship to authorship-attribution approaches.

Background: diachronic Biblical Hebrew

The CBH/LBH distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel-Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 10]. At the lexical level, the first-person singular pronoun shifts from *ankī* (CBH) to *anī* (LBH); the relative pronoun *ašer* gives way to the enclitic *š*e-; the negator *lō'* is supplemented by and eventually partly replaced by *'ēn*. At the morphological level, the wayyiqtol (narrative imperfect) declines in frequency while the *hāyāh* + participle periphrastic progressive increases. At the syntactic level, infinitive-absolute constructions rise, and the overall sentence structure shifts toward the analytic patterns associated with Aramaic influence [11, 12].

The consensus view [1, 13, 14] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption while formally modeling and partially testing the alternatives.

The diachronic versus dialectal debate

Young, Rezetko, and Ehrensvärd [11] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (Dead Sea Scrolls, ∼150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern interpreted as literary register rather than temporal reality.

Hurvitz and the diachronic mainstream [10] counter that the dialectal alternatives require additional auxiliary hypotheses; and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

Crucially for the present paper: rather than adjudicating this debate purely on a priori grounds, the framework operationalizes it. The *resistant model* tests whether syntactic templating patterns that fall below the level of deliberate stylistic choice agree with the full-model (lexical + morphological + syntactic) date. Systematic disagreement — the full model older than the resistant model — is a falsifiable indicator of archaism.

The circularity problem

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. We adopt three safeguards.

First, the training corpus is restricted to prophetic books with *independent* historical anchoring via synchronisms with datable rulers in Assyrian, Babylonian, and Persian king-lists. Torah texts are excluded from training entirely, so all Torah date estimates are genuine out-of-sample predictions.

Second, a held-out validation set (oracle Jeremiah chapters) was withheld from all feature selection and training. Its correct recovery by the model (Section 9) provides empirical evidence against the circularity objection.

Third, the prior sensitivity analysis (Section 6) explicitly quantifies how much of each HB-VI date is determined by the scholar-specified prior versus the linguistic data.

Beyond these three safeguards, the choice of the prophetic corpus as the training base deserves explicit justification, and the *direction* of any residual misdating error is methodologically important.

Why the prophetic corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — "the fifth year of King Jehoiachin's captivity" (Ezek 1:2), "the sixth year" (8:1), and so forth — which can be correlated with Babylonian administrative texts that independently date Jehoiachin's deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are independently documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates are derived entirely from literary-critical arguments, which is precisely the kind of circularity the present study seeks to avoid. The prophetic corpus therefore represents the most defensible choice for a training set whose dates are genuinely independent of the linguistic features under analysis.

Asymmetric error and conservative upper limits. If the accepted dates for the prophetic training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference (the superscriptions and Ezekielian dates provide firm termini post quos), but they could in principle be *younger* — composed or reaching final form some time after the events they describe. This asymmetry has a direct consequence for the date estimates produced by the model. A training corpus that is misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis; unknown texts dated by reference to that calibration would also be estimated as too early, meaning their true dates would be *later* than the model reports. In other words, if the prophetic training dates carry any systematic error, that error propagates in a single direction: all model-derived dates shift toward greater recency, not greater antiquity. The model's output for a text such as the Song of the Sea — already placed firmly in the Persian period by the resistant model — would be pushed later still if the training corpus were found to be younger than traditionally assumed. The dates reported in this study should therefore be understood as *conservative upper bounds on antiquity*: a text dated here to 400 BCE may be more recent, but the linguistic evidence as calibrated against the prophetic corpus provides no support for a date earlier than that estimate.

Prior computational work: authorship attribution versus diachronic dating

Faigenbaum-Golovin et al. [5] provide the most methodologically rigorous prior computational treatment of the Hebrew Bible. Their Higher Criticism (HC) discrepancy method [5] tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null hypothesis that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, DtrH, and P, the method achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating-lemma lists that directly engage traditional philological categories. The statistical optimality properties of the HC statistic (sensitivity to rare but consistent frequency deviations across a large vocabulary) make it well-suited to its task.

The present framework is designed to answer a different question, and five methodological consequences follow directly from that distinction.

Continuous date posteriors versus categorical corpus assignment. The HC method assigns each text to the most likely of three corpora. Because D and DtrH are both dated to the late monarchic or exilic period, two texts composed at 650 and 590 BCE receive identical labels if both score closest to DtrH; no intra-school temporal resolution is possible. The MLE-MVN and HB-VI models instead compute a posterior probability distribution over a 500-point continuous date grid, recovering whatever temporal resolution the data support and quantifying uncertainty explicitly via credible intervals.

Morphosyntactic features versus raw lemma frequencies. Faigenbaum-Golovin et al. operate on raw lemma and lemma n-gram frequencies — a bag-of-words representation in which content words (e.g., *gold* in Tabernacle chapters, *king* in regnal narratives) dominate the signal. Content-word frequencies are sensitive to *topic* and *genre* as much as to *authorship*, and they carry little temporal signal for cross-genre comparisons. The 36 features used in this study are drawn from a century of diachronic Hebrew linguistics scholarship [1–3, 11] precisely because they are sensitive to *period* rather than *topic*: the *ankī*→*anī* pronoun shift, the negator transition, the wayyiqtol decline, and the infinitive-absolute rate change systematically across time and resist topical or generic influence. This distinction is especially important for cross-genre comparisons: bag-of-words features cannot compare poetic or legal texts against narrative texts without large genre confounds.

Archaism diagnostic. The HC method provides no mechanism for distinguishing a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes vocabulary to D/DtrH norms will be assigned to DtrH regardless of actual composition date. The resistant/full-model divergence introduced in Section 7 detects this pattern directly: surface features under conscious compositional control (lexicon, morphology) and deep clause-level syntactic patterns that operate below deliberate stylistic choice are evaluated independently, and systematic divergence between their date estimates constitutes a quantified, falsifiable indicator of archaism. This diagnostic has no counterpart in the HC framework.

External holdout validation. Faigenbaum-Golovin et al. validate via leave-one-out and k-fold cross-validation within the three training corpora. This confirms internal consistency but cannot test whether the method recovers *correct* calendar dates, because the ground-truth labels are scribal-school assignments rather than independently anchored absolute dates. The present framework is validated against texts anchored to external historical events — oracle Jeremiah chapters to the Neo-Babylonian period [15], Haggai to 520 BCE (Persian regnal date), Daniel Hebrew chapters to 167 BCE (Maccabean crisis) — providing a genuinely independent test that the temporal signal is real rather than self-referential.

Cross-linguistic generalization. The Greek validation corpus (63 texts, five held-out texts spanning Archaic through Byzantine Greek) demonstrates that the methodology captures a general diachronic morphosyntactic signal and does not exploit properties idiosyncratic to Biblical Hebrew. No cross-linguistic generalization of the HC authorship method has been reported.

These five differences are differences of purpose rather than deficiencies in either approach. Authorship attribution and diachronic dating ask related but separable questions; the ideal analysis of any disputed text would apply both. The HC output (scribal-school membership) can in principle be used to set register priors in HB-VI, and conversely HB-VI date posteriors constrain the plausible compositional settings for HC attribution decisions. We return to this complementarity in the Discussion.

Data: corpus and feature extraction

The BHSA database

All Hebrew text analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database, version 2021, produced by the Eep Talstra Centre for Bible and Computer (ETCBC) at Vrije Universiteit Amsterdam [6, 7, 16]. BHSA provides morphological tagging at the word level, phrase-level function annotation, clause boundary segmentation, and lexeme coding for the entire Hebrew Bible. Data are accessed programmatically via the Text-Fabric Python library, enabling reproducible extraction without manual annotation. No Aramaic chapters are included in training or testing.

Training corpus

The training corpus comprises 22 prophetic and post-exilic prose units spanning 760–167 BCE (Table 1). Date anchoring uses explicit synchronisms with datable ancient Near Eastern events: for example, Amos 1:1 places the book in the reigns of Uzziah and Jeroboam II (∼760 BCE); Haggai 1:1 gives an explicit Persian regnal date (520 BCE); Daniel's Hebrew chapters (1, 8–12) are anchored to 167 BCE by the Maccabean crisis.

Several calibration decisions require note. Isaiah 56–66 (Trito-Isaiah) is set at 450 BCE, reflecting scholarly near-consensus for Persian-period composition. Ezra and Nehemiah are set at 350 BCE, reflecting arguments for late-Persian or early Hellenistic final form. Daniel is restricted to Hebrew chapters 1 and 8–12; the Aramaic chapters (2–7) are a different language and are excluded.

Oracle Jeremiah chapters (the oracular stratum: chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51) and the Deuteronomistic prose stratum (chs. 7, 11, 17–18, 21, 24–29, 32–45, 52) are both excluded from training and used exclusively as held-out validation units.

Table 1. Training corpus: 22 units with anchored dates and word counts. Date (BCE) is the scholarly consensus midpoint used as the training label. σ is the uncertainty estimate fed to the HB-VI prior. Register: S = SBH, T = Transitional, L = LBH.

Unit	Register	Date (BCE)	σ (yr)	Words
Amos	S	760	20	2,042
Hosea	S	745	20	2,971
Micah	S	725	20	1,786
Isaiah 1–39	S	720	30	8,241
Nahum	S	650	20	618
Habakkuk (training)	S	605	20	670
Zephaniah	S	630	20	1,009
Jeremiah oracle	S	605	30	11,842
Ezekiel	T	580	20	18,793
Isaiah 40–55	T	545	20	4,951
Isaiah 56–66	T	450	30	2,808
Obadiah	T	580	30	291
Joel	T	400	50	1,060
Jonah	T	400	50	687
Malachi	T	460	20	913
Zechariah 1–8	T	518	10	2,302
Zechariah 9–14	L	350	50	1,538
Ezra	L	350	30	4,092
Nehemiah	L	350	30	5,784
Esther	L	300	50	3,122
Chronicles	L	330	30	25,946
Daniel (Heb.)	L	167	10	3,041

Feature extraction

Features are extracted at three tiers that differ in their susceptibility to deliberate stylistic manipulation (Table 2).

Tier 1 (lexical, 20 features). Rates per 1,000 words of specific function words, particles, demonstratives, and prepositions. Examples: the *ašer*/*še* ratio; the *ankī*/*anī* ratio; the absolute rate of the particle *gam*; the rate of the accusative marker *et*. Tier 1 features most directly capture the CBH/LBH lexical axis but are also most susceptible to deliberate archaism, because a competent composer can deliberately choose archaic function words.

Tier 2 (morphological, 12 features). Verb stem fractions (qal, piel, hif, niphal as fractions of all verbs); verb form fractions (wayyiqtol, qatal, yiqtol, participle, imperative fractions); absolute noun ratio; pronominal suffix rate. These are harder to archaize wholesale but remain subject to word-level editorial manipulation.

Tier 3 (clause/phrase, 4 features). Four features forming the *resistant model*: the infinitive-construct fraction (`frac_infc`), the fronted-clause fraction (`frac_fronted`), the null-subject fraction (`frac_null_subj`), and the waw-qatal fraction (`frac_wqtl_wayq`). These measure syntactic templating patterns that are acquired below the level of conscious stylistic choice and are therefore most resistant to deliberate manipulation.

Feature selection. Spearman rank correlation with date is computed for each candidate feature against the 22-text training set. Features are retained at a permissive significance threshold of $p < 0.30$: with only $N = 22$ training texts, a strict Benjamini-Hochberg FDR correction at $\alpha = 0.05$ would retain only $\sim$10 features, which is insufficient to capture the multi-dimensional CBH/LBH signal documented over a century of Hebrew linguistics scholarship. The $p < 0.30$ threshold is paired with a leave-one-out (LOO) robustness filter: a feature is retained only if its Spearman ρ preserves sign in $\geq 68\%$ of LOO resamplings (the 68% threshold corresponds to one standard deviation of a binomial sign-consistency test, providing a principled stopping criterion). In practice, all 35 features surviving the $p < 0.30$ screen achieve 100% LOO sign consistency (LOO fraction = 1.00), well above the 68% floor, confirming that the selected features carry a robust diachronic signal. This yields a final set of **35 features**.

Table 2. Feature tier summary. J = number of features in tier. Susceptibility indicates the degree to which deliberate archaism can manipulate features within that tier.

Tier	Type	J	Archaism susceptibility
1	Lexical (function words, particles)	20	High
2	Morphological (verb forms, stems)	12	Medium
3	Clause/phrase syntax	4	Low (resistant model)
Total selected		35	

Word n-gram model

As an independent grammatical-sequence dating instrument, we extract POS-tag n-gram features that are immune to topic variation and largely immune to deliberate lexical archaism: a composer can choose archaic vocabulary but cannot easily rewrite every grammatical-sequence pattern to match an earlier era.

Two feature types are used. **Type A** features are POS-tag bigrams and trigrams on all words (e.g., `verb_wayyiqtol` · `noun`, `prep` · `prps`), capturing deep grammatical flow. **Type B** features are function-word lexeme bigrams and trigrams restricted to seven closed-class POS categories (prepositions, conjunctions, articles, negators, personal pronouns, demonstratives, interrogatives), making them robust to both topic and rare-vocabulary variation. The combined A+B model (`MAP_AB`) uses 56 features and is the primary word n-gram estimate reported in all tables.

Character n-gram model: assessment and exclusion. An earlier version of the framework included character 3-gram and 4-gram frequencies. Reliability testing on 1QS (Community Rule, Dead Sea Scrolls; independently dated ~150 BCE) returned MAP = 760 BCE — a 600-year error attributable to deliberate orthographic archaism. Sub-morphemic patterns are not immune to deliberate archaism in the Biblical Hebrew corpus. The character n-gram model is therefore excluded from all analyses; this negative result is reported because it is informative about the limits of the approach.

The MLE-MVN model

Per-feature OLS regression

Each feature f_j ($j = 1, \ldots, J$, $J = 35$) is modeled as a linear function of date d plus Gaussian noise:

$$f_j(d) = \alpha_j + \beta_j\, d + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Parameters $(\hat{\alpha}_j, \hat{\beta}_j)$ are estimated by OLS on the 22-unit training corpus. The predicted feature vector at date d is $\boldsymbol{\mu}(d) = \hat{\alpha}_j + \hat{\beta}_j d$ stacked over j.

The linear model is a simplification: some features (e.g., wayyiqtol fraction) may have mild non-linear trends. A sensitivity analysis with quadratic terms shows negligible improvement over the date range 760–167 BCE, justifying the linear approximation.

Tikhonov-regularized residual covariance

The $J \times J$ residual covariance matrix $\mathbf{\Sigma}$ captures co-variation among features unexplained by the date trend. Because the training corpus has only $N = 22$ units, $\mathbf{\Sigma}$ estimated from the raw residuals is near-singular when

$J = 35$. Tikhonov (ridge) regularization is applied:

$$\mathbf{\Sigma}_{\text{reg}} = \mathbf{\Sigma} + \lambda \cdot \frac{\text{tr}(\mathbf{\Sigma})}{J}\, \mathbf{I}_J, \tag{2}$$

where $\lambda = 0.10$ for the full model and $\lambda = 0.20$ for the resistant model (four features only). A sensitivity analysis varying $\lambda \in [0.05, 0.30]$ shows MAP estimates stable to ± 30 yr across representative training and holdout texts. As expected, 68% credible interval widths increase monotonically with λ, growing by 50–70 yr over the tested range (from $\sim$110–135 yr at $\lambda = 0.05$ to $\sim$160–200 yr at $\lambda = 0.30$), reflecting the trade-off between regularization stability and posterior confidence. The canonical value $\lambda = 0.10$ was chosen to balance near-singular covariance suppression against excessive CI widening.

Posterior computation

A 500-point date grid spanning 900 BCE to 100 CE (step ≈ 2 yr) is evaluated. The log-likelihood of observed feature vector $\mathbf{x}$ at grid point d is

$$\ell(d) = -\tfrac{1}{2}\big(\mathbf{x} - \boldsymbol{\mu}(d)\big)^{\top} \mathbf{\Sigma}_{\text{reg}}^{-1} \big(\mathbf{x} - \boldsymbol{\mu}(d)\big). \tag{3}$$

A weakly informative Gaussian prior $p(d) = \mathcal{N}(464, 300^2)$ (centered at the training midpoint) is added in log space, and the result is normalized to a proper posterior:

$$p(d \mid \mathbf{x}) \;\propto\; \exp\big\{\ell(d)\big\} \cdot p(d). \tag{4}$$

The MAP estimate is the mode; 68% and 95% highest-density credible intervals (HDCIs) are extracted numerically.

The HB-VI model

Motivation and limitations of MLE-MVN

The MLE-MVN model treats each feature regression independently and uses point estimates for all parameters. Two failure modes motivate the hierarchical extension. First, the resulting 68% CIs are extremely narrow (median 3–4 yr for SBH texts), indicating the model is systematically overconfident; parameter uncertainty in $\hat{\beta}_j$ is not propagated. Second, register assignment is hard: a text must be pre-assigned to one of three groups (SBH/Transitional/LBH) before the posterior is computed. Misassignment cascades into catastrophically wrong dates (e.g., Haggai assigned to LBH: MAP = 361 BCE vs. 520 BCE actual; error = 159 yr; holdout MAE for MLE-MVN = 134 yr).

Hierarchical model architecture

The HB-VI model is a hierarchical Bayesian regression in which per-group feature loadings are drawn from a global distribution:

$$\beta_{r,j} \sim \mathcal{N}(\mu_{\beta_j}, \, \sigma_{\beta}^2), \tag{5}$$

where $r \in \{1, 2, 3\}$ indexes the register group (SBH, Transitional, LBH), j indexes the feature, and μ_{β_j} and σ_β are global hyperparameters shared across groups. Groups share statistical strength via the global hyperparameters while maintaining group-specific slopes. Intercepts $\alpha_{r,j}$, noise variance $\sigma_{r,j}$, and the global hyperparameters are all variational parameters.

The training corpus for the HB-VI model uses 19 texts (7 SBH, 6 Transitional, 6 LBH), with the three holdout texts (Habakkuk, Haggai, Daniel) excluded, and the full Jeremiah book replaced by the oracle Jeremiah chapters as the SBH anchor. Date priors for training texts are $\mathcal{N}(d_{\text{BCE}}, \sigma_{\text{date}}^2)$, directly encoding known date uncertainty.

Variational inference training

The posterior $p(\boldsymbol{\theta} \mid \mathbf{X})$ is approximated by mean-field variational inference (MFVI): the variational family $q(\boldsymbol{\theta}) = \prod_i q(\theta_i)$ factorizes over all parameters. The Evidence Lower Bound (ELBO) is maximized:

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X} \mid \boldsymbol{\theta})] - \mathrm{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta})) \,. \tag{6}$$

Optimization uses ADAM [8] (learning rate = 0.02, 3,000 iterations) with the reparameterization trick for gradient flow through Gaussian sampling. Convergence is declared when the ELBO improvement over 100 iterations falls below 0.01%; in practice convergence occurs within ∼2,000 iterations. The implementation uses the `autograd` library [17] to avoid GPU dependency.

Date posteriors and best-group selection

For a text with known register, the date posterior is computed by evaluating the group-specific log-likelihood over a 1,000-point BCE grid, integrating over parameter uncertainty via Monte Carlo ($N_{\mathrm{MC}} = 200$ samples from the variational posterior).

For texts with unknown register (documentary sources D, P, JE; short sub-sources), *best-group selection* is used: the marginal log-likelihood is computed for all three groups via `logsumexp` over the date grid, and the group maximizing marginal log-likelihood is selected. This avoids the hard pre-assignment failure mode of MLE-MVN.

Training ceiling effect. The HB-VI model, like MLE-MVN, cannot reliably extrapolate beyond the training range (∼760–330 BCE). Archaic texts whose features cluster with early SBH are mapped to the training ceiling (∼760 BCE), and the model can only report "consistent with the oldest SBH anchor," not a pre-monarchic date. This limitation is acknowledged in all results involving archaic poetry.

Sub-source results

Nine sub-source texts are extracted from BHSA via chapter-range-based selection and dated using the fitted HB-VI parameters (no retraining required). Table 3 reports MAP dates under two prior regimes: Mode A (scholarly prior) and Mode B (agnostic prior; see Section on Prior Sensitivity Analysis).

Table 3. HB-VI sub-source and source MAP dates (BCE). Mode A uses scholarly priors $\mathcal{N}(d, \sigma^2)$; Mode B uses an agnostic prior $\mathcal{N}(575, 400^2)$. Shift = Mode A − Mode B. Classification: $|\text{shift}| < 30$ yr = data-driven (D); 30–80 yr = mildly prior-influenced (M); >80 yr = prior-dominated (P). P_source is the Priestly documentary source composite (Lev and selected Gen/Exo/Num chapters); it is listed here because it is cited as data-driven in the text but does not correspond to a single extractable sub-source range.

Sub-source	Mode A	68% CI	Mode B	Shift	Class
Jer_oracle	637	[659–615]	660	−27	D
Jer_DTR	614	[636–593]	543	+51	M
D_Code	698	[722–673]	737	−51	M
D_Frame	716	[739–693]	714	−20	D
D_Song	797	[831–760]	781	−58	M
Lev_Holiness	687	[715–658]	660	−46	M
Lev_Priestly	646	[676–616]	674	−47	M
P_source	402	[369–436]	406	−4	D
Song_Sea	768	[802–732]	691	+21	D
Song_Deborah	821	[851–792]	741	−34	M

Prior sensitivity analysis

The prior-domination problem

When the date prior $\mathcal{N}(d_{\mathrm{BCE}}, \sigma^2)$ has small σ, the posterior MAP is dominated by the prior rather than the linguistic data. This is especially dangerous when the prior encodes the scholarly consensus dates being tested: the model would then be recovering the prior's opinion, not the linguistic evidence. We introduce a three-mode sensitivity analysis to classify results.

Three prior modes

Mode A (scholarly prior): standard run; σ from the scholarly metadata (5–300 yr depending on text).

Mode B (agnostic prior): $\mathcal{N}(575, 400^2)$ for all texts; covers the training range nearly uniformly, with $\pm 2\sigma$ spanning 225–975 BCE.

Mode C (likelihood-only): $\sigma_n = 10$ (near-delta noise), effectively computing the mode of the raw likelihood; no prior influence.

The shift metric is $\Delta_{AB} = \mathrm{MAP}(A) - \mathrm{MAP}(B)$. Texts are classified as data-driven ($|\Delta_{AB}| < 30$ yr), mildly prior-influenced (30–80 yr), or prior-dominated (> 80 yr).

Results

Of the ten entries in Table 3, four are data-driven: Song_Sea ($\Delta_{AB} = +21$ yr), P_source (-4 yr), D_Frame (-20 yr), and Jer_oracle (-27 yr). Six are mildly prior-influenced (30–80 yr shifts). The three comparison texts used for model validation (Habakkuk, Haggai, Daniel) are all prior-dominated ($|\Delta_{AB}| > 80$ yr), and the documentary source composites D_source and JE_source are also prior-dominated ($\Delta_{AB} = +178$ and $+172$ yr respectively).

The data-driven result for Song_Sea deserves a caveat. Both Mode A (712 BCE) and Mode B (691 BCE) are near the 760 BCE training ceiling — the oldest anchor point in the corpus. When posteriors pile against the training boundary, the A/B shift will be small regardless of true data informativeness, because both prior regimes are compressed into the same constrained region of the date grid. The small $\Delta_{AB} = 21$ yr is therefore necessary but not sufficient evidence of data-dominance: it is consistent with a genuinely informative archaic linguistic signal, but also consistent with a ceiling artifact. The resistant-model MAP (460 BCE) and the word n-gram MAP (783 BCE) provide independent evidence that the archaic clustering is real, but the absolute date for Song_Sea should be treated as a lower bound ("no older than $\sim$760 BCE in the model") rather than a precise calendar estimate.

The prior-dominated status of the comparison texts does not invalidate the model; it reflects that tight scholarly priors ($\sigma = 5$–20 yr, anchored by historical synchronisms) correctly dominate the posterior when the prior is very precise. The HB-VI MAE improvement over MLE-MVN (16.8 yr vs. 134.2 yr) is attributable to better register assignment, not to more data-informative posteriors for these prior-dominated texts.

Archaism detection: the resistant model

The archaism confound

Deliberate lexical archaism inflates apparent textual age in the full model by pulling Tier 1 and Tier 2 features toward the CBH end. Conversely, transmission-with-updating (scribal modernization of vocabulary in a genuinely old document) deflates apparent age. Both produce a systematic divergence between lexical/morphological and clause-syntactic feature signals.

Resistant model

The resistant model uses only the four Tier 3 clause/phrase features (Section on Feature Extraction), with higher regularization ($\lambda = 0.20$). The resulting posterior is wider but less contaminated by lexical confounds.

The full/resistant MAP diagnostic

The *archaism index* is defined as $\Delta_{\text{arch}} = \text{MAP}_{\text{full}} - \text{MAP}_{\text{resist}}$. Positive Δ_{arch} (full model dates older than resistant) indicates archaizing composition; negative values indicate transmission-with-updating.

Representative results illustrate both patterns. For Song of the Sea (Exod 15): $\text{MAP}_{\text{full}} = 852$ BCE, $\text{MAP}_{\text{resist}} = 460$ BCE, $\Delta_{\text{arch}} = +392$ yr — the largest positive value in the dataset. The word n-gram model ($\text{MAP}_{\text{AB}} = 783$ BCE) also dates the text archaically, but the clause-level syntax (resistant model) and the pattern of all three models in mutual agreement are the key evidence for the archaizing interpretation (Fig 4). For D_Code (Deut 12–26): $\text{MAP}_{\text{full}} = 292$ BCE, $\text{MAP}_{\text{resist}} = 847$ BCE, $\Delta_{\text{arch}} = -555$ yr — the most extreme negative value — consistent with a genuinely old legal corpus whose vocabulary has undergone post-exilic updating.

LBH archaism score

Each feature is normalized to a $[0, 1]$ scale:

$$s_j = \frac{x_j - \hat{f}_j(d_{\text{CBH}})}{\hat{f}_j(d_{\text{LBH}}) - \hat{f}_j(d_{\text{CBH}})}, \tag{7}$$

where $d_{\text{CBH}} = 720$ BCE and $d_{\text{LBH}} = 250$ BCE. The mean score $\bar{s}$ across features classifies texts: $\bar{s} < 0.30 =$ Archaic/SBH; 0.30–0.65 = Transitional; $> 0.65 =$ Modern/LBH; $\bar{s} < 0 =$ Archaizing (more CBH-like than any training text at the CBH anchor date). Song of the Sea returns $\bar{s} = -0.325$ (hyper-archaic); P source returns $\bar{s} = 0.731$ (LBH-Modern). The ordering D ($\bar{s} = 0.41$) < JE ($\bar{s} = 0.65$) < P ($\bar{s} = 0.73$) by mean LBH score is the most consistent cross-method signal in the dataset, where lower score indicates more archaic (CBH-like) morphosyntax. This ordering places D as the most archaic source, which may appear to contradict the standard Documentary Hypothesis chronology (oldest to youngest: JE $\rightarrow$ D $\rightarrow$ P). The explanation is two-fold. First, D's legal code (D_Code, $\bar{s} = 0.28$) is heavily archaizing in formulaic diction and vocabulary, pulling D's aggregate score below JE's. Second, the JE composite as extracted from Genesis–Numbers has undergone substantial post-exilic scribal updating that raises its overall LBH score relative to any hypothetical J/E original. The D < JE < P ordering is therefore consistent with a combination of deliberate archaism in D's legal core and differential transmission updating in JE, not with D being linguistically younger than JE.

Hard-terminus features

Certain features function as independent chronological constraints, providing hard termini that can cross-check or override model-derived posteriors. (We deliberately avoid the term "leakage features," which in machine-learning contexts denotes test-set information contaminating training — a different concept entirely.) The complete absence of the enclitic relative pronoun *še-* (rate = 0.00) in Deuteronomy and the Holiness Code sets a terminus ante quem: neither text can be late-Persian or Hellenistic, since *še-* is a robust LBH marker that appears with high frequency in texts of the 5th century and later. Conversely, shared vocabulary or grammatical constructions with independently datable compositions provides terminus post quem estimates. These hard-terminus constraints are used to sanity-check probabilistic results before interpretation.

Genre correction

The genre contamination problem

The training corpus is ∼85% prophetic literature. Legal texts (Deuteronomy, Leviticus) use formulaic vocabulary and specific clause-patterns (heavy *kī* usage, participial law formulations) that resemble either early or late material independently of composition date. Without correction, the model interprets legal register as a temporal signal.

A genre-neutral sub-model — using only features whose between-genre variance (legal vs. narrative within Torah) is $< 20\%$ of temporal variance across training units — reveals the scale of contamination. For Deuteronomy whole, the full-model MAP is 292 BCE while the genre-neutral MAP is 728 BCE, a 436-year discrepancy; the full-model date for D is dominated by genre, not time. For P source, the two models agree to within 58 yr, confirming P's late date is robust to genre correction.

Strategy B: genre-feature down-weighting

Within-Torah feature variation is treated as genre noise. A genre ratio per feature j is:

$$\gamma_j = \frac{\mathrm{Var}(\text{legal} - \text{narrative})_j}{\mathrm{Var}_{\text{temporal},j}}, \tag{8}$$

and the feature weight is $w_j = 1/(1+\gamma_j)$. The genre-weighted log-likelihood is

$$\ell_B(d) = -\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}(d))^\top \mathbf{W}\, \boldsymbol{\Sigma}_{\text{reg}}^{-1} \mathbf{W}\,(\mathbf{x} - \boldsymbol{\mu}(d)), \tag{9}$$

where $\mathbf{W} = \mathrm{diag}(w_j)$.

Strategy D: covariance inflation

Mismatch levels quantify how far each genre class sits from the prophetic training baseline. The training corpus is ∼85% prophetic, so prophetic texts receive mismatch = 0.0 (no inflation needed). The mean genre ratio $\bar{\gamma}$ averaged over all selected features is approximately twice as large for legal texts as for narrative texts within the Torah, motivating a 2:1 weighting: legal = 1.0, narrative = 0.5, prophetic = 0.0. These three levels are the coarsest taxonomy justified by the small training corpus; sub-genre distinctions (apodictic vs. casuistic law) require a larger corpus and are deferred to future work. Extra diagonal variance is then added: $\Sigma_{jj} \mathrel{+}= \text{mismatch} \cdot \gamma_j \cdot \Sigma_{jj}$. This widens the posterior for genre-mismatched texts without shifting the MAP.

Both corrections are applied simultaneously (B+D) for all non-prophetic texts. The mean correction for legal texts is +22 yr (genre correction moves legal-text dates slightly older); for narrative, +12 yr.

Validation

Oracle Jeremiah: primary held-out test

The oracular chapters of Jeremiah (chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51) were excluded from all feature selection and model training. The MLE-MVN full model applied to the oracle chapters returns MAP = 562 BCE, 68% CI = [587–537 BCE], mean LBH score = 0.16 (Archaic/SBH-like). This is consistent with the historical context: Jeremiah was active ca. 627–580 BCE and the oracle chapters preserve his 7th-century prophetic material. The model correctly assigns a held-out text to the correct century without any prior knowledge of its date (Fig 3).

Leave-one-out cross-validation

LOO cross-validation refits the model on 21 units and predicts the withheld unit's date. LOO MAP dates are within ±30–50 yr of full-model MAP dates for all 22 training units, confirming the model is not severely over-fit to any single text. The LOO fraction (proportion of features whose temporal coefficient sign is preserved in LOO fits) is used as a per-feature robustness criterion; features below 65% LOO retention are excluded during feature selection.

Kings/Chronicles parallel validation

Five matched narrative pairs from Samuel-Kings and Chronicles (none in training) are analyzed: the Solomon narrative, selected Judean king narratives, the Hezekiah pericope (2 Kgs 18–20 / Isa 36–39 shared verbatim), the Manasseh-to-Exile narrative, and the Kings-vs.-Isaiah comparison. The word n-gram delta (Kings minus Chronicles date) is +143 to +168 BCE per unit, confirming the model detects that Samuel-Kings is systematically older than its Chronicles parallel in grammatical-sequence patterns.

The Hezekiah cross-check (2 Kgs 18–20 vs. Isa 36–39, near-identical text) yields a word n-gram delta of +9 BCE, establishing the noise floor below 10 yr on effectively identical text.

HB-VI vs. MLE-MVN: head-to-head comparison

Training-set note. The two models use different training sets, and this distinction matters for interpreting Table 4. The MLE-MVN model is trained on all 22 corpus units in Table 1, which includes Habakkuk and Daniel but not Haggai (Haggai was omitted from the MLE-MVN training set because its small word count, 670 words, makes covariance estimation unreliable in a 35-feature space). The HB-VI model is trained on the 19-text subset that excludes all three comparison texts (Habakkuk, Haggai, and Daniel). A consequence is that the MLE-MVN column in Table 4 is not a genuine out-of-sample estimate for Habakkuk and Daniel: those two texts were seen during MLE-MVN training. The HB-VI column is a genuine out-of-sample estimate for all three. For Haggai — the one text unseen by both models — the MLE-MVN error is 159 yr and the HB-VI error is 2 yr, illustrating the register-assignment failure in isolation.

Three comparison texts with independent historical anchoring are used: Habakkuk (605 BCE), Haggai (520 BCE), and Daniel Hebrew (167 BCE). Results are summarized in Table 4.

Table 4. Holdout validation: HB-VI versus MLE-MVN. Scholarly date is the independently anchored target. All dates in BCE.

Text	Scholarly	HB-VI MAP	MLE-MVN MAP	HB-VI error
Habakkuk	605	640	580	+35 yr
Haggai	520	522	361	+2 yr
Daniel	167	180	386	+13 yr
MAE		**16.8 yr**	**134.2 yr**	

The MAE improvement for HB-VI is driven by correct register assignment: Haggai is correctly classified as Transitional (MAP = 522 BCE) rather than miscategorized as LBH (MAP = 361 BCE as in MLE-MVN); Daniel is correctly classified as LBH/late-Transitional (MAP = 180 BCE) rather than being misassigned. This illustrates the core failure mode of the hard register pre-assignment in MLE-MVN: a single wrong classification propagates into a catastrophically wrong date.

Statistical caveat. With only $n = 3$ comparison texts, the standard error of the MAE is large. For HB-VI, individual absolute errors are 35, 2, and 13 yr (mean 16.7 yr, SE $\approx$ 10 yr). For MLE-MVN, they are 25, 159, and 219 yr (mean 134.3 yr, SE $\approx$ 57 yr). Although the 95% intervals on the two MAEs do not overlap, the small sample size means these figures should be interpreted qualitatively: the comparison demonstrates that HB-VI's soft register assignment avoids the catastrophic misclassification errors of MLE-MVN, not that the precise MAE ratio is reliable.

CI width comparison: MLE-MVN 68% CIs are extremely narrow (median 3–4 yr for SBH and LBH texts) and under-cover (0% empirical coverage for holdouts in the nominal 68% CI). HB-VI CIs are wider (median 33–55 yr) and achieve approximately nominal 67% empirical coverage for training texts (Fig 7).

Cross-language validation: Ancient Greek

The identical MLE-MVN framework is applied to a 63-text Ancient Greek corpus spanning $\sim$450 BCE to 300 CE (S4 Appendix), with four register classes (ancient Attic, Atticizing, Koine, Septuagint/LXX). The corpus includes classical prose authors (Thucydides, Plato, Demosthenes), Hellenistic historians and philosophers (Polybius, Diodorus, Philo), New Testament texts, Septuagint books, and Byzantine authors through Diogenes Laërtius, providing a $\sim$750-year span for calibration. Thirty-eight features analogous to the Hebrew set are extracted: optative and infinitive rates (morphological), particle frequencies (*men*, *de*, *gar*, *oun*, *kai*), sentence-initial conjunction rate (the Semitic-substrate diagnostic), and function-word frequencies for closed-class POS categories. The canonical ridge parameter $\lambda = 0.10$ is used; the 500-point date grid spans 500 BCE to 500 CE. Five texts are held out as blind validation: Polybius (−160 BCE), Mark (+70 CE), Matthew (+85 CE), Luke (+120 CE), and Diogenes Laërtius (+230 CE). All five recover MAP dates within ∼30 yr of established scholarly dates (Fig 9).

This result has two implications. First, the core framework — OLS regression per feature, Tikhonov regularization, resistant-model archaism detection — generalizes to a language with very different morphological and syntactic profiles. Second, an analogous LXX/Semitic-substrate diagnostic (high sentence-initial *καί* rate, high *εἶπεν* rate, near-zero *μέν* rate) correctly identifies Mark and Luke as LXX-register texts despite their Greek surface, a direct structural parallel to the archaism detection problem in Hebrew.

Discussion

Limitations

Small training corpus. $N = 22$ dateable texts is small relative to $J = 36$ features, necessitating Tikhonov regularization and yielding wider credible intervals than would be achievable with a larger corpus. The model is calibrated to the available data; no texts with secure independent dates suitable for training are being withheld.

Residual circularity. Training dates are anchored historically but are not independent of scholarly tradition; for texts anchored only to broad regnal periods, the training date carries uncertainty that cannot be fully propagated. The prior sensitivity analysis (Section on Prior Sensitivity Analysis) partially quantifies this source of circularity for HB-VI results.

Oral prehistory. The framework dates the written linguistic surface. Oral traditions preserved in written texts — possible for archaic poetry, legal formulae, or tribal lore — are invisible to the model; dates represent the written composition, not any hypothetical prior oral form.

Coarse genre correction. Genre is modeled at three levels (prophetic, narrative, legal). Sub-genre distinctions (apodictic vs. casuistic law; etiological vs. court narrative) require a larger corpus before they can be treated as distinct classes.

Word n-gram limitations for short texts. Texts under ∼1,000 words yield noisy POS-tag n-gram frequencies. Short units (Obadiah: 291 words; Haggai: 670 words) are flagged and results treated as indicative.

Extensibility

The framework is language-agnostic: any corpus with independent date anchoring, extractable surface features, and a register taxonomy can be analyzed. Promising near-term applications include:

- Qumran sectarian texts (1QS, CD, 1QM) — testing the archaizing-register hypothesis quantitatively.
- Akkadian administrative archives — Old Babylonian versus Neo-Babylonian feature shifts in cuneiform corpora.
- Greek papyri from Oxyrhynchus — diachronic analysis spanning Ptolemaic through Byzantine periods.

The resistant-model concept is portable: any language's most syntactically conservative features can be isolated as the resistant-model input; the archaism diagnostic follows automatically.

Relationship to authorship-attribution approaches

The present framework is designed to complement rather than supersede authorship-attribution methods such as that of Faigenbaum-Golovin et al. [5]. Their Higher Criticism discrepancy method excels at assigning disputed passages to known scribal schools and at identifying the specific lemmas driving that assignment, grounding computational results in categories already meaningful to philologists. The present framework excels at recovering calendar-year estimates with calibrated uncertainty, detecting archaism via the resistant/full divergence, and correcting for genre confounds through Strategies B+D.

A productive joint analysis would proceed in two stages. First, apply the HC method to determine the most likely scribal milieu; this directly informs the register prior in HB-VI — for example, a strong HC assignment to D/DtrH would motivate a SBH register prior, while a P assignment would motivate a Transitional or LBH prior. Second, apply the HB-VI diachronic model with that informed prior to obtain a date posterior and archaism index. Such a two-stage pipeline would combine the interpretability and school-assignment precision of the HC approach with the temporal resolution, uncertainty quantification, and archaism sensitivity of the present framework. Conversely, the present framework's date posteriors constrain the plausible compositional settings for HC attribution decisions, since a text whose resistant model places at 460 BCE is unlikely to represent D material regardless of lexical surface similarity. Developing this joint pipeline is a natural direction for future work.

Conclusion

We have presented a fully probabilistic framework for diachronic dating of ancient texts, comprising two complementary Bayesian models (MLE-MVN and HB-VI), a principled archaism diagnostic (resistant model), a genre-correction framework (Strategies B+D), and an independent grammatical-sequence dating instrument (word n-gram model).

The key empirical results are: (1) oracle Jeremiah, held out from all training, is correctly assigned to the 7th century; (2) HB-VI achieves holdout MAE of 16.8 yr versus 134.2 yr for MLE-MVN, with the improvement driven by soft register assignment; (3) prior sensitivity analysis identifies four entries as genuinely data-driven (Song_Sea, P_source, D_Frame, Jer_oracle), with the Song_Sea result confirming a genuine archaic linguistic profile regardless of the prior; (4) cross-language validation on 63 Ancient Greek texts recovers all five holdout dates within $\sim$30 yr, demonstrating language-agnosticism.

The negative result — the exclusion of the character n-gram model after its failure on 1QS (MAP = 760 BCE vs. expected $\sim$150 BCE) — is reported because it is methodologically important: sub-morphemic orthographic patterns cannot be assumed immune to deliberate archaism.

Companion publications apply the framework to the Pentateuchal documentary sources (D, P, JE), the Song of the Sea (Exod 15), and the sub-units of Deuteronomy and Leviticus, using the method established here.

Supporting information

S1 Appendix. Feature extraction pipeline and BHSA query code. Complete Python scripts for extracting all 36 morphosyntactic features and 56 word n-gram features from the BHSA/Text-Fabric database, including feature definitions, normalization procedures, and chapter-range selectors for sub-source extraction.

S2 Appendix. HB-VI model implementation. Complete `autograd`-based Python implementation of the hierarchical Bayesian variational inference model, ADAM training loop, ELBO convergence diagnostics, and date-posterior computation.

S3 Appendix. Prior sensitivity analysis: full results table. Complete Mode A / Mode B / Mode C MAP dates and 68% credible intervals for all 24 analyzed texts, with Δ_{AB} shift values and classification.

S4 Appendix. Ancient Greek corpus manifest. Complete list of 63 Ancient Greek training texts with author, work, date (CE), register class assignment, and word count. Texts are drawn from the Perseus Digital Library [18] and Thesaurus Linguae Graecae [19].

S1 Table. Full feature set. All 36 selected morphosyntactic features with Spearman ρ, p-value (BH-corrected), LOO retention fraction, genre ratio γ_j, and tier assignment.

S2 Table. Word n-gram feature set. All 56 selected Type A and Type B n-gram features with Spearman ρ and LOO retention fraction.

Data availability

All feature extraction scripts, Bayesian dating models, prior sensitivity code, and corpus manifests are publicly available at `https://github.com/aaronadair/diachronic-hebrew` (to be finalized at time of submission). The underlying Hebrew text corpus is the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA), version 2021, available at `https://github.com/ETCBC/bhsa` under a Creative Commons Attribution 4.0 license [6]. The Ancient Greek corpus is drawn from the Perseus Digital Library (`http://www.perseus.tufts.edu`) and the Thesaurus Linguae Graecae (`http://stephanus.tlg.uci.edu`), both publicly accessible. No proprietary data were used.

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Fig 1. Training corpus timeline and full-model date posteriors. Each of the 22 training units is plotted along the time axis (BCE) with posterior curves computed by leave-one-out cross-validation. Dot size indicates word count. Shaded bands mark the SBH, Transitional, and LBH zones defined by mean LBH score thresholds. Oracle Jeremiah (held-out validation text) is shown as an open circle; its posterior (MAP = 562 BCE) falls squarely in the SBH zone.

Fig 2. Top-10 morphosyntactic features by Spearman $|\rho|$ with date. Each panel shows the rate of one feature (y-axis) versus date BCE (x-axis) across the 22 training units, with OLS regression line and 95% confidence band. Features are color-coded by tier (Tier 1 = lexical, Tier 2 = morphological, Tier 3 = clause/phrase resistant). The *ankī/anī* ratio and wayyiqtol fraction show the steepest monotonic declines; infinitive-construct fraction (Tier 3) shows the most scatter but is retained as a resistant-model feature.

Fig 3. Oracle Jeremiah validation: full and resistant model posteriors. Both the full model (36 features) and the resistant model (4 Tier-3 features) are applied to the oracle Jeremiah chapters, which were excluded from training. Shaded bands indicate 68% and 95% HDCIs. The full-model MAP (562 BCE) and the resistant-model MAP (538 BCE) agree within 24 yr, confirming the oracle chapters carry no archaizing signal. The Deuteronomistic prose stratum (Jer_DTR, also held out) is shown for comparison.

Fig 4. Full versus resistant MAP dates: archaism diagnostic scatter plot. Each of 23 analyzed textual units is plotted with full-model MAP (x-axis) versus resistant-model MAP (y-axis). The diagonal reference line indicates concordance (full = resistant). Points above the diagonal (full older than resistant) indicate archaizing composition; points below indicate transmission-with-updating. Song of the Sea (Exod 15) and D_Code (Deut 12–26) are labeled as the most extreme positive and negative outliers, respectively.

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Fig 5. Genre sensitivity of the 36 morphosyntactic features. Features are ranked by genre ratio γ_j (between-genre variance as a fraction of temporal variance). Features to the right of the dashed threshold ($\gamma_j > 0.20$) are down-weighted in Strategy B genre correction. The 12 features selected for the genre-neutral sub-model are highlighted; these carry primarily temporal signal.

Fig 6. Genre correction (Strategy B+D): full model versus corrected MAP dates. Each bar shows the full-model MAP (gray) and B+D-corrected MAP (color) for Torah books and selected non-prophetic texts. Legal texts (Deuteronomy, Leviticus) shift older under correction; narrative texts (Genesis, Numbers) shift modestly; prophetic texts are near-zero. The rightmost panel compares genre-neutral MAP (open circles) to B+D MAP for legal texts, showing the large genre contamination for D and the convergence for P.

Fig 7. HB-VI versus MLE-MVN: MAP dates and 68% credible intervals for all training and holdout texts. Connected pairs of dots show MLE-MVN (triangles) and HB-VI (circles) MAP dates for each text; vertical bars indicate 68% HDCIs. The holdout texts (Habakkuk, Haggai, Daniel; shaded region) illustrate the key improvement: HB-VI correctly assigns Haggai to the Transitional group (MAP = 522 BCE) while MLE-MVN miscategorizes it as LBH (MAP = 361 BCE). HB-VI CIs are systematically wider and better-calibrated than MLE-MVN CIs.

Fig 8. Prior sensitivity analysis: Mode A versus Mode B MAP dates for all sub-sources and holdouts. Mode A (scholarly prior, circles) and Mode B (agnostic prior, triangles) MAP dates are connected for each text. Line length represents the Δ_{AB} shift; texts are colored by classification: green = data-driven ($|\Delta_{AB}|<30$ yr), orange = mildly prior-influenced (30–80 yr), red = prior-dominated (> 80 yr). The four data-driven texts (Song_Sea, P_source, D_Frame, Jer_oracle) are annotated.

Fig 9. Cross-language validation on Ancient Greek (five holdouts). Each holdout is plotted at its independently established date (x-axis) versus the MLE-MVN MAP date (y-axis) with 68% CI whiskers. All five holdouts fall within ∼30 yr of the diagonal. Register class (ancient Attic, Koine, LXX, Atticizing) is indicated by symbol shape. The inset shows the LXX register diagnostic features for Mark and Luke, confirming the Semitic-substrate signal is correctly isolated from the temporal signal.

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